



DIRE DAWA UNIVERSITY

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Dire Dawa,
Ethiopia

FINAL INTERNSHIP TECHNICAL REPORT (SENG-4182)

Architecting & Engineering a High-Concurrency Multi-Vendor Digital Raffle Platform with Double- Entry Accounting and Provably Fair Cryptography

Project: LuckyEthio Enterprise Infrastructure

STUDENT NAME

Enter Student Full Name

STUDENT ID

DDU/UGR/XXXX/XX

HOST ORGANIZATION

SORARDI PLC (Technology & Software Solu

INTERNSHIP PERIOD

Sene 01, 2017 E.C. – Meskerem 05, 2018 E.C

ACADEMIC ADVISOR

Advisor Name, MSc/PhD

DATE OF SUBMISSION

10/01/2018 E.C.

2. ACKNOWLEDGMENT

First and foremost, I offer my profound thanks to Almighty God for providing me the perseverance, good health, and intellectual clarity to successfully accomplish my Software Engineering internship and author this technical report.

I express my deepest appreciation to **Dire Dawa University, Institute of Technology (IoT), School of Computing, and the Department of Software Engineering** for curating a forward-looking curriculum and facilitating this vital industry attachment. Special gratitude is extended to my **Academic Advisor** for providing continuous academic oversight and constructive evaluation.

I am deeply indebted to **SORARDI PLC**, my **Industry Supervisor**, and the entire software engineering team. Their mentorship, willingness to share advanced system architecture knowledge, and confidence in assigning me core financial, cryptographic, and backend concurrency modules on the LuckyEthio digital platform were instrumental to the success of this internship.

3. LIST OF ACRONYMS & ABBREVIATIONS

ACRONYM	DEFINITION
API	Application Programming Interface
CBE	Commercial Bank of Ethiopia
ECA	Ethiopian Communications Authority
ETB	Ethiopian Birr
HMAC	Hash-based Message Authentication Code
MoR	Ministry of Revenues (Federal Democratic Republic of Ethiopia)

ACRONYM	DEFINITION
NLA	National Lottery Administration (Ethiopia)
PDPP	Personal Data Protection Proclamation (No. 1321/2024)
POS	Point of Sale
RBAC	Role-Based Access Control
RNG	Random Number Generator
SHA	Secure Hash Algorithm (e.g. SHA-256)
USSD	Unstructured Supplementary Service Data (e.g. *157#)
VAT	Value Added Tax (15% Statutory Rate)

5. INTRODUCTION

The Software Engineering Internship (SEng-4182) at Dire Dawa University is a cornerstone practical course designed to immerse students into enterprise software development workflows. The attachment allows students to apply theoretical knowledge from courses like *Database Systems*, *Web Engineering*, *Software Architecture*, and *Computer Security* directly to commercial systems.

During my 4-month attachment with **SORARDI PLC**, I was embedded within the core engineering team responsible for architecting and building high-traffic, secure, and regulatory-compliant digital ticketing and financial settlement systems.

6. HOST COMPANY PROFILE: SORARDI PLC

SORARDI PLC is an innovative Ethiopian software development and technology solutions enterprise specializing in modern web architectures, enterprise resource planning (ERP), digital financial systems, and high-concurrency cloud applications.

Organizational Hierarchy: The engineering division at SORARDI PLC comprises the Chief Technology Officer, Frontend UI/UX Team, Core Systems & Backend Team (where I served as Software Engineering Intern), Quality Assurance, and Regulatory Compliance Officers.

7. INTERNSHIP ACTIVITIES & ROLES

My primary responsibilities at SORARDI PLC centered around full-stack feature engineering, database architecture, and cryptographic protocol implementation:

- **Monorepo Engineering:** Restructured the codebase into a modular Turborepo containing `apps/web`, `apps/admin`, `packages/database`, and `packages/shared`.
- **Double-Entry Financial Ledger:** Engineered balanced ledger transactions (Debits = Credits) for ticket proceeds, 15% VAT escrow, seller holdings, and agent commissions.
- **Atomic Ticket Concurrency:** Implemented database-level transaction locks (`tx.`) and composite unique constraints to guarantee zero overselling under high concurrency.
- **Multi-Provider Payment Adapters:** Built unified provider adapters for Chapa, Telebirr, CBE Birr, and SantimPay with idempotent webhook handlers.
- **Provably Fair RNG Engine:** Designed a multi-entropy commit-reveal formula with immutable `DrawSnapshot` state locks.
- **Data Privacy Governance:** Implemented customer data rights workflows adhering to Ethiopian PDPP Proclamation No. 1321/2024.

8. TECHNICAL SKILLS UTILIZED

DOMAIN	TECHNOLOGIES & METHODOLOGIES APPLIED
Frontend	Next.js 14 (App Router), React 18, Tailwind CSS, Lucide Icons, Responsive Layouts
Backend & API	Node.js, TypeScript, REST API Route Handlers, Signed Webhook Verification

DOMAIN	TECHNOLOGIES & METHODOLOGIES APPLIED
Database & Storage	Prisma ORM, SQLite / PostgreSQL, Relational Schema Normalization
Financial Accounting	Double-Entry General Ledger, Chart of Accounts, Trial Balance Verification
Cryptography & Security	SHA-256 Hashing, HMAC Signatures, Two-Person Authorization, Dynamic QR Tokens
Project Management	Agile Scrum, 2-Week Sprints, GitHub Pull Requests, Static Type Checking

9. COLLABORATION & TEAMWORK

Working within SORARDI PLC's Agile Scrum framework, I actively participated in daily standups, code review sessions, and backlog refinement. Key collaborative milestones included re-engineering the mobile drawer navigation to float on the right-hand side, standardizing dual-consent seller draw agreements, and building the stealth administrative authentication portal.

10. PROJECT DEEP DIVE: LUCKYETHIO PLATFORM HIGHLIGHTS

The platform decouples ticket purchasing into a robust multi-stage pipeline:

```
// Decoupled Lifecycle:
Customer Intent → PurchaseOrder (PUR-XXXX) → Payment Gateway (Chapa/Telebirr)
      ▲
      | (Webhook Verified + Idempotency Guard)
      ▼
Atomic DB Transaction → Allocate Tickets (TKT-XXXX) → Double-Entry Ledger Posting
```

Provably Fair Formula: $Winner = (BigInt(Sha256(v2.0 : RaffleID : SecretSeed : SoldTickets : PublicEntropy : Algorithm)) \% SoldTickets) + 1$

11. CHALLENGES ENCOUNTERED & ENGINEERING SOLUTIONS

CHALLENGE ENCOUNTERED	ROOT CAUSE	ENGINEERING SOLUTION IMPLEMENTED
Inventory Race Conditions	Concurrent requests reading stale counts.	Atomic database transactions with <code>UNIQUE(raffleId, ticketNumber)</code> .
Duplicate Webhooks	Gateway retries sending duplicate HTTP posts.	Enforced strict <code>idempotencyKey</code> tracking.
Ledger Drift	Rounding discrepancies across VAT and escrow.	Strict double-entry journal ensuring <code>Total Debits === Total Credits</code> .

12. ORGANIZATIONAL CONTRIBUTIONS

I delivered 57 fully compiled Next.js routes (34 web pages, 23 admin pages) with **zero TypeScript errors**, configured Turborepo caching for lightning-fast builds, and authored complete regulatory and user documentation for SORARDI PLC.

13. CONCLUSION & RECOMMENDATIONS

This internship at SORARDI PLC successfully bridged software engineering theory with the rigorous demands of production fintech and digital gaming platforms.

- **For SORARDI PLC:** Implement hardware security modules (HSM) for automated secret seed rotation and integrate telecom SMPP gateways.
- **For Dire Dawa University:** Incorporate enterprise monorepo tooling (Turborepo) and financial ledger design into advanced programming labs.

14. REFERENCES

1. National Lottery Administration (NLA) of Ethiopia, *Proclamation No. 535/2007*.
2. Ethiopian Communications Authority (ECA), *Personal Data Protection Proclamation No. 1321/2024*.
3. Ministry of Revenues (MoR), *Value Added Tax (VAT) Proclamation No. 285/2002*.
4. Vercel & Next.js Core Team, *Next.js 14 App Router Documentation*, 2024.
5. Prisma ORM Team, *Prisma Client & Concurrency Documentation*, 2024.

15. TECHNICAL APPENDICES

Appendix A: Database Schema Summary (18 Models)

Models: Role, Permission, RolePermission, User, Seller, Agent, AgentLedger, AgentAccessLog, Raffle, PurchaseOrder, PaymentAttempt, Ticket, Transaction, LedgerAccount, LedgerTransaction, LedgerEntry, TaxLedger, CashoutRequest, DrawSnapshot, DrawAudit, TwoPersonApproval, AuditLog, RiskEvent, DataSubjectRequest.